Q1 - 25 July - Shift 1

An observer is riding on a bicycle and moving towards a hill at 18 kmh^{-1}. He hears a sound from a source at some distance behind him directly as well as after its reflection from the hill. If the original frequency of the sound as emitted by source is 640 Hz and velocity of the sound in air is 320 m/s, the beat frequency between the two sounds heard by observer will be _____ Hz.

Space for your notes:

Q2 - 26 July - Shift 1

When a car is approaching the observer, the frequency of horn is 100Hz. After passing the observer, it is 50Hz. If the observer moves with the car, the frequency will be $\frac{x}{3}$ Hz where x = ___

Space for your notes:

Q3 - 26 July - Shift 2

A transverse wave is represented by $y = 2\sin(\omega t - kx)$ cm. The value of wavelength (in cm) for which the wave velocity becomes equal to the maximum particle velocity, will be ;

(A) 4π (B) 2π

(C) π (D) 2

Space for your notes:

Q4 - 27 July - Shift 2

A wire of length 30 cm, stretched between rigid supports, has it's n^{th} and $(n+1)^{th}$ harmonics at 400 Hz and 450 Hz, respectively. If tension in the string is 2700 N, it's linear mass density is……..kg/m.

Space for your notes:

Q5 - 28 July - Shift 1

In the wave equation

$$y = 0.5 \sin \frac{2\pi}{\lambda}(400t - x) m$$

the velocity of the wave will be :

(A) 200 m/s (B) $200\sqrt{2}$ m / s

(C) 400 m/s (D) $400\sqrt{2}$ m / s

Space for your notes:

Q6 - 28 July - Shift 1

The frequency of echo will be _________ Hz if the train blowing a whistle of frequency 320 Hz is moving with a velocity of 36 km/h towards a hill from which an echo is heard by the train driver. Velocity of sound in air is 330 m/s.

Space for your notes:

Q7 - 29 July - Shift 1

Two light beams of intensities 4I and 9I interfere on a screen. The phase difference between these beams on the screen at point A is zero and at point B is π. The difference of resultant intensities, at the point A and B, will be _________ I.

Space for your notes:

Q8 - 29 July - Shift 2

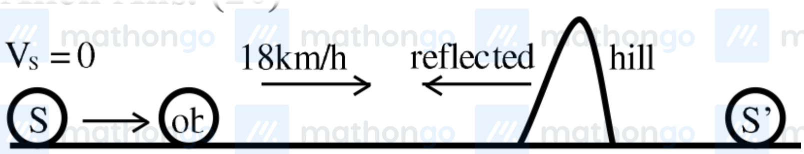
The speed of a transverse wave passing through a string of length 50 cm and mass 10 g is 60 ms^{-1}. The area of cross-section of the wire is 2.0 mm^2 and its Young's modulus is 1.2×10^{11} Nm^{-2}. The extension of the wire over its natural length due to its tension will be $x \times 10^{-5}$ m. The value of x is_______.

Space for your notes:

Answer Key

Q1 (20)	**Q2** (200)	**Q3** (A)	**Q4** (3)
Q5 (C)	**Q6** (340)	**Q7** (24)	**Q8** (15)

Q1 (20)



$$V_S = 0,\ V_{ob} = 5m/s$$

$$f_{direct} = \left(\frac{320-5}{320}\right)640 = 630Hz$$

$$f_{reflected} = \left(\frac{320+5}{320}\right)640 = 650Hz$$

$$f_{beat} = 650 - 630 = 20Hz$$

Q2 (200)

$$f_1 = 100 = f_0\left(\frac{C}{C-V_s}\right)$$

C = speed of sound

V_S = speed of source

$$f_2 = 50 = f_0\left(\frac{C}{C+V_s}\right)$$

$$\frac{f_1}{f_2} = 2 = \frac{C + V_s}{C - V_s}$$

$$2C - 2V_s = C + V_s$$

$$3V_s = C$$

$$V_s = \frac{C}{3}$$

$$100 = f_0 \frac{C}{\frac{2C}{3}} = \frac{3}{2} f_0$$

$$f_0 = \frac{200}{3}$$

Q3 (A)

$y = 2\sin(\omega t - kx)$

Maximum particle velocity $= A\omega$

Wave velocity $= \dfrac{\omega}{k}$

$$\frac{\omega}{k} = A\omega$$

$$k = \frac{1}{A} = \frac{2\pi}{\lambda}$$

$$\lambda = 2\pi A$$

$$= 4\pi \text{ cm}$$

Q4 (3)

$$\frac{nv}{0.6} = 400 \ \& \ \frac{(n+1)v}{0.6} = 450$$

$$\Rightarrow \left[\frac{0.6 \times 400}{v} + 1\right]\frac{v}{0.6} = 450$$

$$\Rightarrow = v = 30$$

$$\Rightarrow \sqrt{\frac{T}{\mu}} = 30$$

$$\Rightarrow \frac{2700}{\mu} = 900 = \mu = 3$$

Q5 (C)

$$y = 0.5\sin\left(\frac{2\pi}{\lambda}400t - \frac{2\pi}{\lambda}x\right)$$

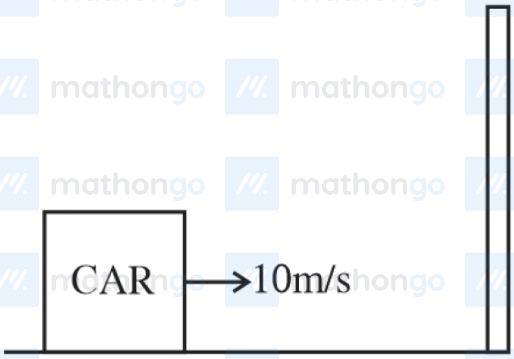
$$\omega = \frac{2\pi}{\lambda}400$$

$$K = \frac{2\pi}{\lambda}$$

$$v = \frac{\omega}{k} \qquad [v = 400\ \text{m/s}]$$

Q6 (340)

The hill will be a secondary source.



f_1 = frequency of the car w.r.t. the hill

$$f_1 = \left(\frac{v}{v - v_s}\right)f = \left(\frac{330}{320}\right)\times 320 = 330\,\text{Hz}$$

f_2 = Frequency of the sound reflected by hill w.r.t. the car (echo)

$$f_2 = \left(\frac{v + v_0}{v}\right)f_1 = \frac{(330 + 10)}{330}\times 330 = 340\,\text{Hz}$$

Q7 (24)

$$I_{net} = I_1 + I_2 + 2\sqrt{I_1}\sqrt{I_2}\cos\phi$$

I_{max} for $\phi = 0$ & I_{min} for $\phi = \pi$

$$I_{max} = \left(\sqrt{I_1} + \sqrt{I_2}\right)^2 = \left(\sqrt{9I} + \sqrt{4I}\right)^2 = 25\,I$$

$$I_{min} = \left(\sqrt{I_1} - \sqrt{I_2}\right)^2 = \left(\sqrt{9I} - \sqrt{4I}\right)^2 = I$$

$$I_{max} - I_{min} = 25\,I - I = 24\,I$$

Q8 (15)

$$V_w = \sqrt{\frac{T}{\mu}}$$

$$60 = \sqrt{\frac{T}{10 \times 10^{-3}} \times 0.5}$$

$$T = \frac{(60)^2 \times 10^{-2}}{0.5} = 72\text{ N}$$

$$\Delta\ell = \frac{F\ell}{AY} = \frac{72 \times 0.5}{2 \times 10^{-6} \times 1.2 \times 10^{11}}$$

$$= \frac{72 \times 5}{24} \times 10^{-5} = 15 \times 10^{-5}$$

Ans. 15